

Yr 11 Mathematics - Task 1 (2009)



GIRRAWEE HIGH SCHOOL
YEAR 11 - TASK 1 (2009)
MATHEMATICS

Time allowed – 90 min

- Start each question on a *new* sheet of paper.

Question 1 (16 marks)

- (a) Find $\frac{\sqrt{34 \div 5.9}}{21 - 4.7}$ correct to 3 decimal places. 1
- (b) Evaluate $\sqrt[3]{\frac{73.1}{0.2}}$ correct to 2 significant figures. 1
- (c) Write 0.01072 in scientific notation. 1
- (d) Fully simplify $\sqrt{245}$. 2
- (e) If $\sqrt{12} - \sqrt{3} = x\sqrt{3}$, find the value of x . 2
- (f) Which of the following $\pi, \sqrt{1}, -\sqrt{9}, \sqrt{3}, 2^{-1}, 5^{\frac{1}{2}}, 8^{-\frac{1}{3}}$ are rational numbers. 3
- (g) Express the following as a fraction in simplest form.
- (i) $0.38\dot{4}$ 3
- (ii) $6.\dot{2}4\dot{7}$ 3

Question 2 (24 marks)

- (a) Simplify.
- (i) $\sqrt{50} - \sqrt{32}$ 2
- (ii) $2\sqrt{3} + \sqrt{27} - \sqrt{243}$ 2
- (iii) $\frac{5\sqrt{2} \times 6\sqrt{10}}{3\sqrt{5}}$ 3
- (b) Expand and simplify.
- (i) $4\sqrt{3}(3\sqrt{6} + 2\sqrt{12})$ 2
- (ii) $(5\sqrt{2} - 3)^2$ 3

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(c) Express with a rational denominator.

$$(i) \frac{2\sqrt{3}}{\sqrt{7}} \quad 2$$

$$(ii) \frac{4}{2-\sqrt{5}} \quad 3$$

$$(iii) \frac{\sqrt{2}}{\sqrt{3}-4\sqrt{5}} \quad 3$$

$$(iv) \frac{7\sqrt{5}-1}{7\sqrt{5}+1} \quad 4$$

Question 3 (22 marks)

(a) Expand and simplify

$$(i) 3a^2\left(2a - \frac{b^2}{a}\right) - 5a(b^2 + 3a) \quad 2$$

$$(ii) (3y^2 - 2)(3y^2 + 2) \quad 2$$

$$(iii) (x + 2)(x^2 - 2x + 4) \quad 2$$

$$(iv) (x + 2)(x - 4)^2 \quad 3$$

(b) Factorise

$$(i) 4m^2 - 25 \quad 2$$

$$(ii) x^2 - 4xy - 60y^2 \quad 2$$

$$(iii) 2y^2 + 11y - 21 \quad 3$$

$$(iv) 64r^3 - 27 \quad 3$$

$$(v) 4 - 64a^4 \quad 3$$

Question 4 (19 marks)

(a) Simplify

$$(i) \frac{3c}{4} - \frac{c}{6} \quad 2$$

$$(ii) \frac{1}{x-2} - \frac{1}{x} \quad 3$$

$$(iii) \frac{2}{x+3} + \frac{x}{x-2} \quad 3$$

(b) Simplify

$$(i) \frac{1-x}{x^2-1} \quad 3$$

$$(ii) \frac{x^3 + 4x^2 - 4x - 16}{x^2 + 6x + 8} \quad 4$$

$$(iii) \frac{a^2 + b^2}{a+b} \div \frac{a^4 - b^4}{a^2 + 2ab + b^2} \quad 4$$

Question 5 (20 marks)

(a) Solve

$$(i) \quad 28 = 12x - 5x \quad 2$$

$$(ii) \quad (2x-1)(x+3) = 0 \quad 2$$

$$(iii) \quad \frac{9}{2x} = 4 - \frac{3}{x} \quad 3$$

$$(iv) \quad 4x^2 + 4x - 15 = 0 \quad 4$$

(b) Complete the following

$$x^2 + 14x + \dots = (x + \dots)^2 \quad 2$$

(c) Solve by completing the square

$$x^2 + 4x = 1 \quad 4$$

(d) Solve by using the quadratic formula

$$2x^2 + 4x - 7 = 0 \quad 3$$

Question 6 (13 marks)

(a) Solve the following

(i) $-2x \leq 4$ 1

(ii) $\frac{2x}{3} < x + 2$ 2

(b) Solve simultaneously

(i) $\begin{matrix} x + 2y = 10 \\ x - y = 1 \end{matrix}$ 3

(ii) $\begin{matrix} 5x + 2y = 11 \\ x - 5y = 13 \end{matrix}$ 3

(iii) $\begin{matrix} y = x^2 + 3 \\ y = 4x \end{matrix}$ 4

Question 7 (16 marks)

(a) Indicate whether each of the following relations is a **function** or **not a function**.
State its **domain**.

(i) $x^2 + y^2 = 9$ 2

(ii) $y = \frac{1}{x-3}$ 2

(iii) $y = \sqrt{\frac{1}{4} - x^2}$ 2

(iv) $y = -1 - x^2$ 2

(b) Graph the function $x - 3y - 3 = 0$, including any intercepts. 3

(c) Graph the function $y = x^2 - 10x - 11$, including the vertex and intercepts. 5

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Q1 a) 0.147

①

b) 7.1

①

c) 1.072×10^{-2}

①

d) $\sqrt{245} = \sqrt{49 \times 5}$
 $= 7\sqrt{5}$

②

e) $\sqrt{12} - \sqrt{3} = x\sqrt{3}$

$2\sqrt{3} - \sqrt{3} = x\sqrt{3}$

$1\sqrt{3} = x\sqrt{3}$

$\therefore x = 1$

②

f) RATIONAL

$\sqrt{1}, -\sqrt{9}, 2^{-1}, 8^{-\frac{1}{3}}$

③

g) i) $x = 0.3844...$

$100x = 38.44...$

$1000x = 384.44...$

$\therefore 1000x - 100x = 384 - 38$

$900x = 346$

$x = \frac{346}{900} = \frac{173}{450}$

③

ii) $x = 6.247247...$

$1000x = 6247.247247...$

$1000x - x = 6247 - 6$

$999x = 6241$

$x = \frac{6241}{999}$

50

$6 \frac{247}{999}$

③

Q2 a) i) $\sqrt{50} - \sqrt{32}$

$= 5\sqrt{2} - 4\sqrt{2}$

$= \sqrt{2}$

②

ii) $2\sqrt{3} + \sqrt{27} - \sqrt{243}$

$= 2\sqrt{3} + 3\sqrt{3} - 9\sqrt{3}$

$= -4\sqrt{3}$

②

iii) $\frac{5\sqrt{2} \times 6\sqrt{10}}{3\sqrt{5}}$

$= \frac{30\sqrt{20}}{3\sqrt{5}}$

$= 10\sqrt{\frac{20}{5}}$

$= 10\sqrt{4}$

$= 20$

③

b) i) $4\sqrt{3}(3\sqrt{6} + 2\sqrt{12})$

$= 12\sqrt{18} + 8\sqrt{36}$

$= 36\sqrt{2} + 48$

②

ii) $(5\sqrt{2} - 3)^2$

$= 25(2) - 2(5\sqrt{2})(3) + 9$

$= 59 - 30\sqrt{2}$

③

c) i) $\frac{2\sqrt{3}}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$

$= \frac{2\sqrt{21}}{7}$

②

$$c) ii) \frac{4}{2-\sqrt{5}} \times \frac{(2+\sqrt{5})}{(2+\sqrt{5})}$$

$$= \frac{8+4\sqrt{5}}{4-5}$$

$$= \frac{8+4\sqrt{5}}{-1}$$

$$= -8-4\sqrt{5}$$

(3)

$$iii) \frac{\sqrt{2}}{\sqrt{3}-4\sqrt{5}} \times \frac{\sqrt{3}+4\sqrt{5}}{\sqrt{3}+4\sqrt{5}}$$

$$= \frac{\sqrt{6}+4\sqrt{10}}{3-16(5)}$$

$$= \frac{\sqrt{6}+4\sqrt{10}}{-77}$$

(3)

$$iv) \frac{7\sqrt{5}-1}{7\sqrt{5}+1} \times \frac{7\sqrt{5}-1}{7\sqrt{5}-1}$$

$$= \frac{49(5) - 2(7\sqrt{5})(1) + 1}{49(5) - 1}$$

$$= \frac{246 - 14\sqrt{5}}{244}$$

$$= \frac{123 - 7\sqrt{5}}{122}$$

(4)

$$(Q3) a) i) 3a^2 \left(2a - \frac{b^2}{a} \right) - 5a(b^2 + 3a)$$

$$= 6a^3 - 3ab^2 - 5ab^2 - 15a^2$$

$$= 6a^3 - 8ab^2 - 15a^2$$

$$= a(6a^2 - 8b^2 - 15a)$$

(2)

$$a) ii) (3y^2-2)(3y^2+2)$$

$$= 9y^4 - 4$$

(2)

$$iii) (x+2)(x^2-2x+4)$$

$$= x^3 - 2x^2 + 4x + 2x^2 - 4x + 8$$

$$= x^3 + 8$$

(2)

$$iv) (x+2)(x-4)^2$$

$$= (x+2)(x^2-8x+16)$$

$$= x^3 - 8x^2 + 16x + 2x^2 - 16x + 32$$

$$= x^3 - 6x^2 + 32$$

(3)

$$b) i) 4m^2 - 25$$

$$= (2m-5)(2m+5)$$

(2)

$$ii) x^2 - 4xy - 60y^2$$

$$= (x-10y)(x+6y)$$

(2)

$$iii) 2y^2 + 11y - 21$$

$$= 2y^2 + 14y - 3y - 21$$

$$= 2y(y+7) - 3(y+7)$$

$$= (y+7)(2y-3)$$

(3)

$$iv) 64r^3 - 27$$

$$= (4r-3)(16r^2+12r+9)$$

(3)

$$v) 4 - 64a^4$$

$$= 4(1 - 16a^4)$$

$$= 4(1-4a^2)(1+4a^2)$$

$$= 4(1-2a)(1+2a)(1+4a^2)$$

(3)

$$(Q4) a) i) \frac{3c}{4} - \frac{c}{6}$$

$$= \frac{9c}{12} - \frac{2c}{12}$$

$$= \frac{7c}{12} \quad (2)$$

$$ii) \frac{1}{x-2} - \frac{1}{x}$$

$$= \frac{x}{x(x-2)} - \frac{(x-2)}{x(x-2)}$$

$$= \frac{x - x + 2}{x(x-2)}$$

$$= \frac{2}{x(x-2)} \quad (3)$$

$$iii) \frac{2}{x+3} + \frac{x}{x-2}$$

$$= \frac{2(x-2) + x(x+3)}{(x+3)(x-2)}$$

$$= \frac{2x-4 + x^2+3x}{(x+3)(x-2)}$$

$$= \frac{x^2+5x-4}{(x+3)(x-2)} \quad (3)$$

$$b) i) \frac{1-x}{x^2-1}$$

$$= \frac{-(x-1)}{(x-1)(x+1)}$$

$$= \frac{-1}{x+1} \quad (3)$$

$$ii) \frac{x^3+4x^2-4x-16}{x^2+6x+8}$$

$$= \frac{x^2(x+4) - 4(x+4)}{(x+4)(x+2)}$$

$$= \frac{(x+4)(x^2-4)}{(x+4)(x+2)} = \frac{(x-2)(x+2)}{(x+2)} \quad (4)$$

$$= x-2$$

$$b) iii) \frac{a^2+b^2}{a+b} \div \frac{a^4-b^4}{a^2+2ab+b^2}$$

$$= \frac{a^2+b^2}{a+b} \times \frac{(a+b)(a+b)}{(a^2-b^2)(a^2+b^2)}$$

$$= \frac{(a+b)}{(a-b)(a+b)}$$

$$= \frac{1}{a-b} \quad (4)$$

$$(Q5) a) i) 28 = 12x - 5x$$

$$7x = 28$$

$$x = 4 \quad (2)$$

$$ii) (2x-1)(x+3) = 0$$

$$2x-1=0 \quad \text{or} \quad x+3=0$$

$$2x=1$$

$$x = \frac{1}{2}$$

$$x = -3 \quad (2)$$

$$iii) \frac{9}{2x} = 4 - \frac{3}{x}$$

$$9 = 8x - 6$$

$$15 = 8x$$

$$x = 15/8 \quad \text{or} \quad 1\frac{7}{8} \quad (3)$$

$$iv) 4x^2+4x-15=0 \quad \text{or} \quad 2x \times -3$$

$$4x^2+10x-6x-15=0$$

$$2x(2x+5)-3(2x+5)=0$$

$$(2x+5)(2x-3)=0$$

$$\therefore x = -\frac{5}{2}, x = \frac{3}{2} \quad (4)$$

$$b) x^2+14x + \left(\frac{14}{2}\right)^2 = \left(x + \frac{14}{2}\right)^2$$

$$\therefore (7)^2$$

$$x^2+14x + 49 = (x+7)^2 \quad (7) \quad (2)$$

Q5 c) $x^2 + 4x = 1$

$$x^2 + 4x + 4 = 1 + 4$$

$$(x+2)^2 = 5$$

$$x+2 = \pm\sqrt{5}$$

$$x = -2 \pm \sqrt{5}$$

④

d) $2x^2 + 4x - 7 = 0$

$$\therefore x = \frac{-4 \pm \sqrt{4^2 - 4(2)(-7)}}{2(2)}$$

$$= \frac{-4 \pm \sqrt{16 + 56}}{4}$$

$$= \frac{-4 \pm \sqrt{72}}{4}$$

$$= \frac{-4 \pm 6\sqrt{2}}{4}$$

$$x = \frac{-2 \pm 3\sqrt{2}}{2}$$

③

Q6 a) i) $-2x \leq 4$

$$x \geq -2$$

①

ii) $\frac{2x}{3} < x + 2$

$$2x < 3x + 6$$

$$-6 < x$$

$$x > -6$$

②

b) i) $x + 2y = 10$ ①

$$x - y = 1$$
 ②

① - ② $3y = 9$
 $y = 3$

sub $y = 3$ into ②

$$x - 3 = 1$$

$$x = 4$$

$$\therefore (4, 3)$$

③

ii) $5x + 2y = 11$ ①

$$x - 5y = 13$$
 ②

$$x = 5y + 13$$
 ③

sub ③ into ①

$$5(5y + 13) + 2y = 11$$

$$25y + 65 + 2y = 11$$

$$27y = -54$$

$$y = -2$$

sub $y = -2$ into ②

$$x - 5(-2) = 13$$

$$x + 10 = 13$$

$$x = 3$$

$$(3, -2)$$
 ③

Q6) b) iii) $y = x^2 + 3$ ①
 $y = 4x$ ②

sub ① into ② $4x = x^2 + 3$
 $x^2 - 4x + 3 = 0$
 $(x-3)(x-1) = 0$
 $\therefore x = 1, 3$

sub $x=1$ into ②

$y = 4(1)$
 $y = 4$

$\therefore (1, 4)$

sub $x=3$ into ②

$y = 4(3)$
 $y = 12$

$(3, 12)$ ④

c) $y = x^2 - 10x - 11$

$y = (x-11)(x+1)$

(y-int)

$y = 0^2 - 10(0) - 11$
 $y = -11$

(x-int.)

$0 = (x-11)(x+1)$

$\therefore x = 11, -1$

Axis of symmetry

$x = \frac{-b}{2a}$

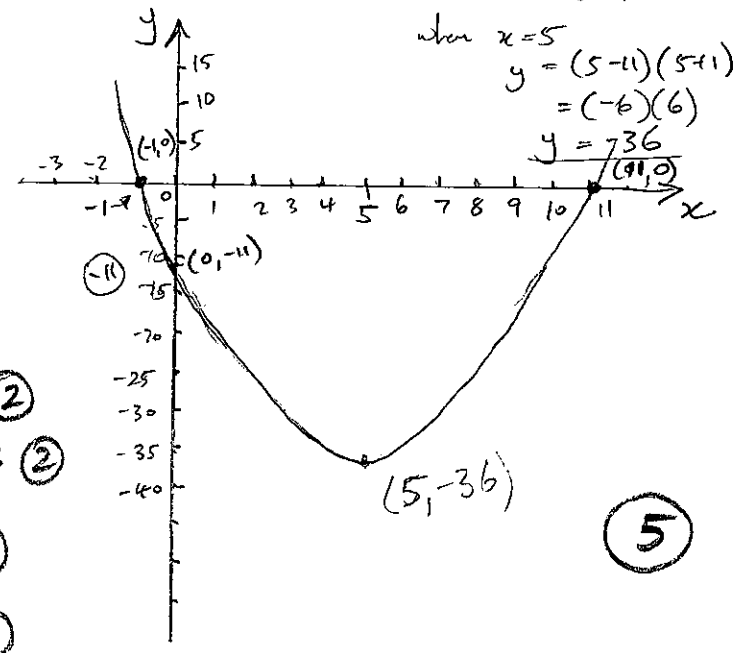
$x = \frac{-(-10)}{2(1)}$

$x = 5$

or Avg. of x-int.

when $x=5$

$y = (5-11)(5+1)$
 $= (-6)(6)$
 $y = -36$



- Q7) a) i) NOT a function; $D: -3 \leq x \leq 3$ ②
 ii) Function; $D: \text{all real } x, x \neq 3$ ②
 iii) Function; $D: -\frac{1}{2} \leq x \leq \frac{1}{2}$ ②
 iv) Function; $D: \text{all real } x$ ②

b) $x - 3y - 3 = 0$

$3y = x - 3$

$y = \frac{1}{3}x - 1$

(y-int)
 $y = \frac{1}{3}(0) - 1$
 $y = -1$

(x-int)
 $0 = \frac{1}{3}x - 1$
 $1 = \frac{1}{3}x$
 $x = 3$

